

Chapter 24: Probability Basics

Overview

Probability (Xác suất) is the mathematical study of randomness and uncertainty. It provides a framework for quantifying how likely events are to occur, forming the foundation for statistics, decision-making, and scientific reasoning.

What you'll master:

- **Sample Spaces and Events:** The fundamental language of probability
- **Probability Axioms:** Basic rules governing probability calculations
- **Compound Events:** Union, intersection, and complement of events
- **Addition Rule:** Computing probabilities of "or" events
- **Multiplication Rule:** Computing probabilities of "and" events
- **Conditional Probability:** Probability with given information
- **Independence:** When events don't influence each other

Real-world connections:

- Medical diagnosis and treatment effectiveness
- Risk assessment in insurance and finance
- Quality control in manufacturing
- Weather forecasting and climate modeling
- Game theory and strategic decision-making

Vocabulary & Terminology

Core Terms

English	IPA	Vietnamese	Definition
Probability	/ˌprɒbəˈbɪləti/	Xác suất	A measure of the likelihood of an event
Sample space	/'sɑːmpəl speɪs/	Không gian mẫu	The set of all possible outcomes

Event	/ɪˈvent/	Biến cố	A subset of the sample space
Outcome	/ˈaʊtkʌm/	Kết quả	A single possible result of an experiment
Random experiment	/ˈrændəm ɪk ˈsperɪmənt/	Phép thử ngẫu nhiên	An action with uncertain outcomes
Certain event	/ˈsɜːtən ɪˈvent/	Biến cố chắc chắn	An event that always occurs (Ω)
Impossible event	/ɪmˈpɒsəbəl ɪˈvent/	Biến cố không thể	An event that never occurs ($\emptyset$)
Complementary event	/ˌkɒmplɪˈmentəri ɪˈvent/	Biến cố đối	The event that A does not occur ($\overline{A}$)
Mutually exclusive	/ˈmjuːtʃuəli ɪk ˈskluːsɪv/	Xung khắc	Events that cannot occur simultaneously
Favorable outcome	/ˈfeɪvərəbəl ˈaʊtkʌm/	Kết quả thuận lợi	An outcome in which the event occurs

Advanced Terms

English	IPA	Vietnamese	Definition
Union of events	/ˈjuːniən əv ɪˈvents/	Biến cố hợp	Event A or B occurs ($A \cup B$)

Intersection of events	/ˌɪntəˈsekʃən əv ɪˈvents/	Biến cố giao	Both events A and B occur ($A \cap B$)
Conditional probability	/kənˈdɪʃənəl ˌprɒbəˈbɪləti/	Xác suất có điều kiện	Probability of A given B occurred ($P(A B)$)
Independent events	/ˌɪndɪˈpendənt ɪˈvents/	Biến cố độc lập	Events where one doesn't affect the other
Addition rule	/əˈdɪʃən ruːl/	Công thức cộng	Formula for $P(A \cup B)$
Multiplication rule	/ˌmʌltɪplɪˈkeɪʃən ruːl/	Công thức nhân	Formula for $P(A \cap B)$
Law of total probability	/lɔː əv ˈtəʊtəl ˌprɒbəˈbɪləti/	Công thức xác suất toàn phần	Computing probability via partitions

Theory

1. Random Experiments and Sample Spaces

1.1 Random Experiments (Phép thử ngẫu nhiên)

Definition: A **random experiment** is an action or process that:

1. Can be repeated under the same conditions
2. Has a well-defined set of possible outcomes
3. Has an uncertain outcome before the experiment is performed

Examples:

- Tossing a coin
- Rolling a die
- Drawing a card from a deck

- Measuring the lifetime of a light bulb

1.2 Sample Space (Không gian mẫu)

Definition: The **sample space**

Ω

(or

S

) is the set of all possible outcomes of a random experiment.

Examples:

Experiment	Sample Space
Toss a coin	$\Omega = \{H, T\}$
Roll a die	$\Omega = \{1, 2, 3, 4, 5, 6\}$
Toss two coins	$\Omega = \{HH, HT, TH, TT\}$
Draw a card	$\Omega = \{\text{all 52 cards}\}$

Notation:

$n(\Omega)$

or

$|\Omega|$

denotes the number of outcomes in the sample space.

1.3 Events (Biến cố)

Definition: An **event** is a subset of the sample space. An event occurs if the outcome of the experiment belongs to that subset.

Types of events:

1. **Certain event** (Biến cố chắc chắn):

Ω

(always occurs)

2. **Impossible event** (Biến cố không thể) :

$\emptyset$

(never occurs)

- 3. **Elementary event** (Biến cố sơ cấp): Contains exactly one outcome
- 4. **Compound event** (Biến cố phức hợp): Contains multiple outcomes

Example: Rolling a die (

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

)

- Event A: "Roll an even number" = $\{2, 4, 6\}$
- Event B: "Roll a number greater than 4" = $\{5, 6\}$

2. Probability of an Event

2.1 Classical Definition (Định nghĩa cổ điển)

For a random experiment with equally likely outcomes:

$$P(A) = \frac{n(A)}{n(\Omega)} = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

Conditions: All outcomes must be equally likely.

2.2 Axioms of Probability (Tiên đề xác suất)

For any event

A

:

1. Non-negativity:

$$0 \leq P(A) \leq 1$$

2. Certainty:

$$P(\Omega) = 1$$

3. Impossibility:

$$P(\emptyset) = 0$$

4. Complement rule:

$$P(\overline{A}) = 1 - P(A)$$

3. Operations on Events

3.1 Complement of an Event (Biến cố đối)

Definition: The **complement** of event A, denoted

$$\bar{A}$$

or

$$A^c$$

, is the event that A does not occur.

Formula:

$$P(\bar{A}) = 1 - P(A)$$

Example: If

$$P(\text{rain}) = 0.3$$

, then

$$P(\text{no rain}) = 1 - 0.3 = 0.7$$

3.2 Union of Events (Biến cố hợp)

Definition: The **union**

$$A \cup B$$

is the event that A occurs OR B occurs (or both).

Venn diagram: The shaded region covers both A and B.

3.3 Intersection of Events (Biến cố giao)

Definition: The **intersection**

$$A \cap B$$

(also written

$$AB$$

) is the event that both A AND B occur.

Venn diagram: The shaded region is where A and B overlap.

3.4 Mutually Exclusive Events (Biến cố xung khắc)

Definition: Events A and B are **mutually exclusive** (or disjoint) if they cannot occur simultaneously:

$$A \cap B = \emptyset$$

Example: When rolling a die:

- A: "Roll an even number" = $\{2, 4, 6\}$
 - B: "Roll a number divisible by 5" = $\{5\}$
 - $A \cap B = \emptyset$
 $\rightarrow$ A and B are mutually exclusive
-

4. Addition Rule (Công thức cộng xác suất)

4.1 Addition Rule for Mutually Exclusive Events

If A and B are mutually exclusive (

$$A \cap B = \emptyset$$

):

$$P(A \cup B) = P(A) + P(B)$$

Generalization: For mutually exclusive events

$$A_1, A_2, \dots, A_n$$

:

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$$

4.2 General Addition Rule

For any two events A and B:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Intuition: We subtract

$$P(A \cap B)$$

because it's counted twice in

$$P(A) + P(B)$$

.

Special case: If

$$A \cap B = \emptyset$$

, then

$$P(A \cap B) = 0$$

, reducing to the formula for mutually exclusive events.

5. Conditional Probability (Xác suất có điều kiện)

5.1 Definition

Definition: The **conditional probability** of A given that B has occurred is:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Interpretation: The probability of A, computed in the reduced sample space where B is known to have occurred.

Example: Drawing cards without replacement:

$$\begin{aligned} &\bullet P(\text{2nd card is Ace} \mid \text{1st card is Ace}) = \\ &\quad \frac{3}{51} \end{aligned}$$

5.2 Properties

1. $0 \leq P(A|B) \leq 1$
 2. $P(\Omega|B) = 1$
 3. $P(\bar{A}|B) = 1 - P(A|B)$
-

6. Multiplication Rule (Công thức nhân xác suất)

6.1 General Multiplication Rule

For any two events A and B:

$$P(A \cap B) = P(B) \cdot P(A|B) = P(A) \cdot P(B|A)$$

Application: Computing probabilities of sequential events.

6.2 Tree Diagrams

Tree diagrams provide a visual method for computing probabilities of sequential events using the multiplication rule.

Structure:

- Each branch represents a stage of the experiment
 - Branch probabilities are conditional probabilities
 - Multiply along paths to get joint probabilities
-

7. Independent Events (Biến cố độc lập)

7.1 Definition

Definition: Events A and B are **independent** if the occurrence of one does not affect the probability of the other:

$$P(A|B) = P(A) \quad \text{and} \quad P(B|A) = P(B)$$

Equivalent condition:

$$P(A \cap B) = P(A) \cdot P(B)$$

7.2 Properties

If A and B are independent, then:

1. $\overline{A}$
and B are independent
2. A and $\overline{B}$
are independent
3. $\overline{A}$
and $\overline{B}$
are independent

7.3 Independence vs. Mutual Exclusivity

Important distinction:

- **Mutually exclusive:**

$$A \cap B = \emptyset$$

(cannot both occur)

- **Independent:**

$$P(A \cap B) = P(A) \cdot P(B)$$

(one doesn't affect the other)

Key insight: If A and B are mutually exclusive with

$$P(A) > 0$$

and

$$P(B) > 0$$

, they CANNOT be independent (because if A occurs, B definitely doesn't occur).

8. Summary of Key Formulas

Rule	Formula	Condition
Complement	$P(\bar{A}) = 1 - P(A)$	Always
Addition (exclusive)	$P(A \cup B) = P(A) + P(B)$	$A \cap B = \emptyset$
Addition (general)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Always
Conditional	$P(A B) = \frac{P(A \cap B)}{P(B)}$	$P(B) > 0$
Multiplication (general)	$P(A \cap B) = P(B) \cdot P(A B)$	Always
Multiplication (independent)	$P(A \cap B) = P(A) \cdot P(B)$	A, B independent

💡 Worked Examples

Example 1: Sample Space and Events

Problem: Two coins are tossed. Let A be the event "at least one head appears" and B be the event "both coins show the same face."

- a) Describe the sample space.
- b) List the outcomes in events A and B.
- c) Find

$$P(A)$$

,

$$P(B)$$

, and

$$P(A \cap B)$$

.

Solution:

- a) **Sample space:**

$$\Omega = \{HH, HT, TH, TT\}$$

$$n(\Omega) = 4$$

- b) **Events:**

- $A = \{HH, HT, TH\}$
(at least one H)
- $B = \{HH, TT\}$
(both same)
- $A \cap B = \{HH\}$
(at least one H AND both same)

- c) **Probabilities:**

$$P(A) = \frac{3}{4}, \quad P(B) = \frac{2}{4} = \frac{1}{2}, \quad P(A \cap B) = \frac{1}{4}$$

Answer: a)

$$\Omega = \{HH, HT, TH, TT\}$$

; b)

$$A = \{HH, HT, TH\}$$

,

$$B = \{HH, TT\}$$

; c)

$$P(A) = \frac{3}{4}$$

,

$$P(B) = \frac{1}{2}$$

,

$$P(A \cap B) = \frac{1}{4}$$

Example 2: Complement Rule

Problem: The probability that a student passes an exam is 0.75. What is the probability that the student fails?

Solution:

Let A = "student passes". Then

$\bar{A}$

= "student fails".

Using the complement rule:

$$P(\bar{A}) = 1 - P(A) = 1 - 0.75 = 0.25$$

Answer: 0.25 or 25%

Example 3: Addition Rule (Mutually Exclusive)

Problem: A card is drawn from a standard 52-card deck. What is the probability of drawing a heart OR a spade?

Solution:

Let:

- H = "draw a heart"
- S = "draw a spade"

These events are mutually exclusive (a card cannot be both a heart and a spade).

$$P(H) = \frac{13}{52} = \frac{1}{4}, \quad P(S) = \frac{13}{52} = \frac{1}{4}$$

Using the addition rule for mutually exclusive events:

$$P(H \cup S) = P(H) + P(S) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Answer:

$$\frac{1}{2}$$

or 50%

Example 4: General Addition Rule

Problem: In a class of 30 students, 18 study French, 15 study Spanish, and 8 study both languages. If a student is selected at random, what is the probability the student studies French OR Spanish?

Solution:

Let:

- F = "studies French"
- S = "studies Spanish"

Given:

$$P(F) = \frac{18}{30}, \quad P(S) = \frac{15}{30}, \quad P(F \cap S) = \frac{8}{30}$$

Using the general addition rule:

$$P(F \cup S) = P(F) + P(S) - P(F \cap S) = \frac{18}{30} + \frac{15}{30} - \frac{8}{30} = \frac{25}{30} = \frac{5}{6}$$

Answer:

$$\frac{5}{6} \approx 0.833$$

or about 83.3%

Example 5: Conditional Probability

Problem: A box contains 7 blue pens and 5 black pens. Son draws one pen at random (without replacement), then Tung draws one pen from the remaining 11 pens. What is the probability that Tung draws a blue pen, given that Son drew a black pen?

Solution:

Let:

- A = "Son draws a black pen"
- B = "Tung draws a blue pen"

We need

$$P(B|A)$$

Method 1: By definition

If A occurs (Son drew black), the box now contains 7 blue and 4 black pens (11 total).

$$P(B|A) = \frac{7}{11}$$

Method 2: Using the formula

$$P(A) = \frac{5}{12}$$

$$P(A \cap B) = \frac{5}{12} \times \frac{7}{11} = \frac{35}{132}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{35/132}{5/12} = \frac{35}{132} \times \frac{12}{5} = \frac{7}{11}$$

Answer:

$$\frac{7}{11} \approx 0.636$$

or about 63.6%

Example 6: Multiplication Rule

Problem: Using the scenario from Example 5, what is the probability that Son draws a black pen AND Tung draws a blue pen?

Solution:

Using the multiplication rule:

$$P(A \cap B) = P(A) \cdot P(B|A) = \frac{5}{12} \times \frac{7}{11} = \frac{35}{132}$$

Answer:

$$\frac{35}{132} \approx 0.265$$

or about 26.5%

Example 7: Independent Events

Problem: A box contains 4 red balls and 5 blue balls. Minh draws one ball, records its color, and returns it to the box. Then Hung draws one ball. Let:

- $A = \text{"Minh draws a red ball"}$
- $B = \text{"Hung draws a blue ball"}$

Show that A and B are independent.

Solution:

Check if

$$P(A \cap B) = P(A) \cdot P(B)$$

:

$$P(A) = \frac{4}{9}$$

Since Minh returns the ball, the composition remains unchanged:

$$P(B) = \frac{5}{9}$$

For the joint event:

$$P(A \cap B) = \frac{4}{9} \times \frac{5}{9} = \frac{20}{81}$$

Also:

$$P(A) \cdot P(B) = \frac{4}{9} \times \frac{5}{9} = \frac{20}{81}$$

Since

$$P(A \cap B) = P(A) \cdot P(B)$$

, events A and B are independent.

Alternative reasoning: The replacement ensures that Minh's draw doesn't affect the composition for Hung's draw, so the events are independent by design.

Answer: A and B are independent.

Example 8: Independence vs. Mutual Exclusivity

Problem: Two dice are rolled. Let:

- $A = \text{"First die shows an even number"}$
- $B = \text{"Second die shows a multiple of 3"}$

a) Are A and B mutually exclusive?

b) Are A and B independent?

Solution:

a) Mutually exclusive?

A and B can occur simultaneously (e.g., first die = 2, second die = 3).

Therefore,

$$A \cap B \neq \emptyset$$

→ **Not mutually exclusive.**

b) Independent?

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

(first die: 2, 4, 6)

$$P(B) = \frac{2}{6} = \frac{1}{3}$$

(second die: 3, 6)

$$P(A \cap B) = P(A) \times P(B) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

Verification: There are 36 equally likely outcomes. Event

$$A \cap B$$

consists of:

• (2,3), (2,6), (4,3), (4,6), (6,3), (6,6) → 6 outcomes

$$P(A \cap B) = \frac{6}{36} = \frac{1}{6}$$

✓

Since

$$P(A \cap B) = P(A) \cdot P(B)$$

, **A and B are independent.**

Answer: a) Not mutually exclusive; b) Independent

Applications

1. Medical Testing and Diagnosis

Problem: A disease affects 1% of the population. A diagnostic test has:

- Sensitivity (true positive rate): 95%
- Specificity (true negative rate): 90%

If a person tests positive, what is the probability they actually have the disease?

Solution:

Let:

- D = "has disease"
- T = "tests positive"

Given:

- $P(D) = 0.01$
, so
 $P(\overline{D}) = 0.99$
- $P(T|D) = 0.95$
(sensitivity)
- $P(\overline{T}|\overline{D}) = 0.90$
, so
 $P(T|\overline{D}) = 0.10$

We need

$$P(D|T)$$

Using Bayes' theorem (derived from multiplication and addition rules):

$$P(D|T) = \frac{P(D) \cdot P(T|D)}{P(D) \cdot P(T|D) + P(\overline{D}) \cdot P(T|\overline{D})}$$

$$= \frac{0.01 \times 0.95}{0.01 \times 0.95 + 0.99 \times 0.10} = \frac{0.0095}{0.0095 + 0.099} = \frac{0.0095}{0.1085} \approx 0.0876$$

Interpretation: Even with a positive test, the probability of actually having the disease is only about 8.76% due to the low base rate (1% prevalence).

2. Quality Control

Problem: A factory produces items on two machines:

- Machine A produces 60% of items, with 2% defective
- Machine B produces 40% of items, with 5% defective

An item is selected at random and found to be defective. What is the probability it came from Machine A?

Solution:

Let:

- A = "from Machine A"
- B = "from Machine B"
- D = "defective"

Given:

- $P(A) = 0.6$
- $P(B) = 0.4$
- $P(D|A) = 0.02$
- $P(D|B) = 0.05$

We need

$$P(A|D)$$

$$P(A|D) = \frac{P(A) \cdot P(D|A)}{P(A) \cdot P(D|A) + P(B) \cdot P(D|B)}$$

$$= \frac{0.6 \times 0.02}{0.6 \times 0.02 + 0.4 \times 0.05} = \frac{0.012}{0.012 + 0.02} = \frac{0.012}{0.032} = 0.375$$

Answer: 37.5% probability the defective item came from Machine A.

3. Traffic Safety Analysis

Problem: A traffic safety study examined 577,006 car accidents:

- Seatbelt worn: 510 deaths, 412,368 survivors
- No seatbelt: 1,601 deaths, 162,527 survivors

Compare the probability of death with and without a seatbelt.

Solution:

Let:

- D = "death in accident"
- S = "seatbelt worn"

Without seatbelt:

$$P(D|\bar{S}) = \frac{1601}{1601+162527} = \frac{1601}{164128} \approx 0.00976$$

With seatbelt:

$$P(D|S) = \frac{510}{510+412368} = \frac{510}{412878} \approx 0.00124$$

Comparison:

$$\frac{P(D|\bar{S})}{P(D|S)} = \frac{0.00976}{0.00124} \approx 7.9$$

Conclusion: Not wearing a seatbelt increases the risk of death in an accident by approximately **7.9 times**.

Summary

Key Concepts

1. **Sample Space:** The set of all possible outcomes (Ω)
2. **Event:** A subset of the sample space
3. **Probability:** A number between 0 and 1 measuring likelihood
4. **Complement:**

$$P(\bar{A}) = 1 - P(A)$$
5. **Union:** A OR B occurs ($A \cup B$)
6. **Intersection:** A AND B occur ($A \cap B$)
7. **Mutually Exclusive:** Cannot both occur ($A \cap B = \emptyset$)
8. **Independent:** One doesn't affect the other ($P(A \cap B) = P(A) \cdot P(B)$)

Essential Formulas

Concept	Formula
Classical probability	$P(A) = \frac{n(A)}{n(\Omega)}$
Complement	$P(\bar{A}) = 1 - P(A)$
Addition (exclusive)	$P(A \cup B) = P(A) + P(B)$
Addition (general)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional	$P(A B) = \frac{P(A \cap B)}{P(B)}$
Multiplication	$P(A \cap B) = P(B) \cdot P(A B)$
Independence	$P(A \cap B) = P(A) \cdot P(B)$

Problem-Solving Strategy

1. **Identify the sample space:** List all possible outcomes
 2. **Define events clearly:** Use set notation
 3. **Determine relationships:** Mutually exclusive? Independent?
 4. **Choose the right formula:** Addition, multiplication, or conditional?
 5. **Check your answer:** Does
 $0 \leq P \leq 1$
? Does it make intuitive sense?
-

Review Questions

Conceptual Questions

1. Explain the difference between mutually exclusive events and independent events.
Can two events be both?
2. Why do we subtract
 $P(A \cap B)$
in the general addition rule?
3. If
 $P(A) = 0.6$
and
 $P(B) = 0.4$
, what can you say about
 $P(A \cup B)$
? What are the minimum and maximum possible values?
4. Give a real-world example of two independent events and two dependent events.

Computational Problems

5. A die is rolled. Find:
 - a) $P(\text{even number})$
 - b) $P(\text{number} > 4)$
 - c) $P(\text{even number OR number} > 4)$
6. Two cards are drawn from a standard deck without replacement. Find the probability that:
 - a) Both are aces
 - b) The first is an ace and the second is a king
 - c) At least one is an ace
7. In a class, 70% of students passed Math, 60% passed Physics, and 50% passed both. If a student is selected at random:
 - a) What is the probability they passed Math OR Physics?
 - b) What is the probability they passed Math given they passed Physics?
 - c) Are passing Math and passing Physics independent?

Application Problems

8. A company has three factories producing the same product:

- Factory 1: 50% of production, 1% defective
- Factory 2: 30% of production, 2% defective
- Factory 3: 20% of production, 3% defective

A randomly selected item is defective. What is the probability it came from Factory 1?

9. A medical test for a disease has 98% sensitivity and 95% specificity. If the disease prevalence is 0.5%, what is the probability a person with a positive test actually has the disease?

10. In a game, you win if you roll a sum of 7 or 11 with two dice. What is the probability of winning?

Memory Aids & Tips

Mnemonics

"OR means ADD, AND means MULTIPLY"

- $P(A \cup B)$
involves addition:
 $P(A) + P(B) - P(A \cap B)$
- $P(A \cap B)$
involves multiplication:
 $P(A) \cdot P(B|A)$

"Independent = Multiply Straight"

- If A and B are independent:
 $P(A \cap B) = P(A) \times P(B)$
(no conditioning needed)

"Complement = 1 Minus"

- $P(\bar{A}) = 1 - P(A)$
(often easier to compute "not A" than "A")

Common Pitfalls

1. Confusing "or" with "and":

- "At least one" usually means OR (union)
- "Both" means AND (intersection)

2. Forgetting to subtract overlap:

- $P(A \cup B) \neq P(A) + P(B)$
unless mutually exclusive

3. Assuming independence:

- Always check if
 $P(A \cap B) = P(A) \cdot P(B)$
before using independence

4. Mixing up

$$P(A|B)$$

and

$$P(B|A)$$

:

- $P(\text{disease} | \text{positive test}) \neq P(\text{positive test} | \text{disease})$

Calculator Tips (Casio fx-580VN X)

Computing probabilities with combinations:

Plain Text

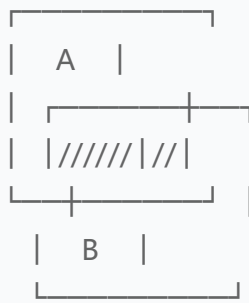
$$P(\text{drawing 2 aces from 52 cards}) = C(4,2) / C(52,2)$$

$$4 \text{ SHIFT } \div (nCr) 2 \div (52 \text{ SHIFT } \div (nCr) 2) =$$
$$\rightarrow 0.00452...$$

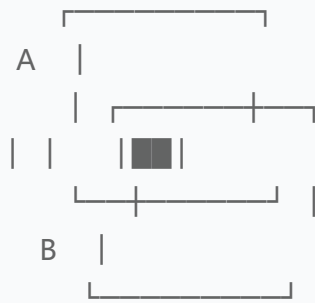
Visual Aid: Venn Diagrams

Plain Text

Union ($A \cup B$):



Intersection ($A \cap B$):



Conclusion

Probability provides the mathematical language for reasoning about uncertainty. The key skills are:

1. Carefully defining sample spaces and events
2. Recognizing relationships between events (exclusive, independent)
3. Selecting and applying the appropriate formula
4. Interpreting results in context

In the next chapter (Chapter 25: Statistics), we'll use probability concepts to analyze data, compute measures of central tendency and dispersion, and make inferences about populations from samples.

Practice recommendation: Work through 15-20 problems covering all formula types, with emphasis on conditional probability and independence, as these concepts are most commonly misunderstood.